

PRIOR — SHARED, UNCHANGED ACROSS ALL PHASES

Score $\mathcal{S} = \{(p_i, b_i, d_i)\}$
known in every phase

Score graph $\rightarrow L_{\mathcal{G}}$
spectral kernel
 $K_G(s) = U g(\nu; s) U^{\top}$

Music model (Phase 0)
per-note embedding h_i

Feature kernels
 $\sum_f \text{diag}(c_f) \otimes X_f X_f^{\top}$
25 score features, h_i

Joint kernel
 $K = B \otimes K_G + \sum_f \text{diag}(c_f) \otimes X_f X_f^{\top} + \text{diag}(\varsigma^2) \otimes I$

Coregionalization B
now couples
 $\tau, \log r, v, c, \text{vibrato}$

$\theta = (s, B, \{c_f\}, \varsigma^2)$
*learned jointly by one
exact per-piece log marginal likelihood*

Exact conjugate
GP posterior
unchanged

Per-note $\tau_i, \log r_i, v_i$
+ c_i , vibrato
each with uncertainty

Monophonic audio
 f_0 extraction

Derived targets
intonation c_i (cents)
vibrato $(f_i^{\text{vib}}, \gamma_i)$

LIKELIHOOD — THE ONLY PART THAT CHANGES

Likelihood
 $\tilde{y} = y + e$ on *observed* notes
 Σ_e now *substantive*: tar-
gets estimated, not exact

Phase 2 (not yet implemented; dashed). Audio enters only to derive per-note targets. The observation model, the prior and the conjugate posterior are unchanged — what changes is the channel set and that the observation noise Σ_e is no longer near-zero.